🗺 The Geological History — FULL MIND MAP

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Chapter 02_viii_The_Geological_History

Topic Index

Ice Ages — Great vs Little

- 'Great Ice Age' = **Pleistocene Epoch** se related (~2 million years ago se ~10,000 years ago tak) — extensive ice sheets/glaciers development ka period. *[Q1]*
- 'Little Ice Age' generally **1650 AD – 1870 AD** ke beech maana jaata hai — Europe/North America mein 20th century se colder winters. *[Q2]*

Dinosaurs ka Period

- Dinosaurs ka period ~**18 crore years pehle** (Jurassic Period, ~20.8 to 14.4 crore years ago) — India mein remains Raiyoli (Gujarat) aur Bara Simla hills (MP) mein mile hain. *[Q3]*

Continental Drift — Tectonic Activities

- Continents apart drift karte hain primarily **Tectonic Activities** ki wajah se (tectonic plates continuously move karti hain). *[Q4]*

Gondwanaland aur Pangaea

- **Pangaea** = originally Earth ka ek hi huge landmass (supercontinent), surrounded by superocean **Panthalassa**. *[Q9]*
- Pangaea split hua **Laurasia (North) aur Gondwanaland (South)** mein. *[Q6]*
- Gondwanaland mein shamil the: South America, Africa, Madagascar, Australia, Antarctica, Arabian Peninsula, Peninsular India — **North America shamil NAHI** tha. *[Q5,Q7]*

⚠ **EXAM TRAP**

• Gondwanaland = Southern landmass, Laurasia = Northern landmass — **North America** Gondwanaland ka part NAHI tha (Laurasia ka tha). *[Q5]*

Plate Tectonics

- **Cocos Plate** Central America aur Pacific Plate ke beech located hai. *[Q8]*
- **Plate Tectonics Theory** — Earth's crust ke pieces mantle ki movement se driven hoke constant slow motion mein rehte hain. 7 major plates: Eurasian, African, Indo-Australian, Pacific, North American, South American, Antarctic. *[Q12]*
- Continental drift ke evidence: Brazil-West Africa coast ki ancient rocks match karti hain (I), Ghana gold deposits Brazilian Plateau se related hain (II), India ka Gondwana sediment system Southern Hemisphere ke 6 landmasses mein match karta hai (III) — teeno statements **correct**. *[Q13]*

Crust Formation aur Destruction

- New crust continuously **Seafloor Spreading** se create hoti hai; old crust **Subduction** se destroy hoti hai — dono balance karte hain, isliye Earth ka size roughly constant rehta hai. *[Q14]*

Folding aur Diastrophism

- **Folding** = **Orogenic force** ka result hai (mountain-building force, tectonic plate interactions se produce hota hai). *[Q15]*
- Diastrophism ke under aate hain: Orogenic process, Earthquakes, Plate tectonics — 'Heterogenic process' naam ki koi standard diastrophism process **nahi** hoti. *[Q16]*

Earth Surface Change — Contributing Factors

- Earth surface pe dynamic changes laane wale factors: Electromagnetic radiation, Geothermal energy, Gravitational force, Plate movements, Earth's rotation, Earth's revolution — **saare 6 factors** relevant hain. *[Q17]*

Prehistoric Formations aur Life Evolution Timeline

- Formations chronological order (oldest to newest): **Sihawal** → **Patpara** → **Baghor** → **Khetaunhi** (Lower Palaeolithic → Middle Palaeolithic → Upper Palaeolithic → Neolithic/Microlithic). *[Q18]*
- Life evolution chronological order: **Shelled animals** → **Insects** → **Reptiles** → **Mammals**. *[Q19]*
- Earliest fossil evidence of life on Earth: exam-key answer **3.5 (million/billion) years ago** (source options have a unit inconsistency, but option B is the accepted key). *[Q10]*
- Continental drift AUR glacial cycles — dono ne organisms ke evolution ko influence kiya hai. *[Q11]*

Thar/Rajasthan Desert — Geological Period

- Rajasthan/Thar Desert **Pleistocene aur recent deposits** ka expanse hai — very low rainfall wala region. *[Q20]*